PAC1

Pràctiques de Procrastinació-Avançada en LaTeX

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I, **Vladimir Scoriae**, declare that I hold the total and full authorship of all the tasks that are carried out in this document. I am the only one who has elaborated each exercise. I have not shared the statements with anyone and the only help I have received has been through the UOC classroom and teachers there.

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1 First UOC section

P^{3}CTEX Sample text with UOC section style.

1.1 UOC subsection

1.1.1 UOC subsubsection

More text with UOC subsection style.
Thin horizontal rule:

Thick horizontal rule (centered):

---

2 pxPRP: property maps and lists

2.1 Objects, keys, and access

`\pxGET␣DEMO.k1[2]` → v2.
`\pxPRINTF␣DEMO.k2[2]` → **DEMO.k2[2]:**vB.

2.2 Overwrite and append

Before overwrite: OVER.state[1] = old. After: OVER.state[1] = newA, OVER.state[2] = newB.

2.3 Lists: `\pxPutValue`, `\pxListSet`, `\pxListAll`

LST.items = (a, b, c).
All values (expected 1–3): 1 – 2 – 3.

2.4 Clear key/object and predicates

Before clear: TMP.k[1] = X. After: TMP.k[1] = Z. Now: TMP.k[1] = P.
Object MISSING is not present, as expected.
CHK exists. Key `nums` exists in CHK. Length of CHK.nums: 3.

2.5 Boolean flags and scalar access

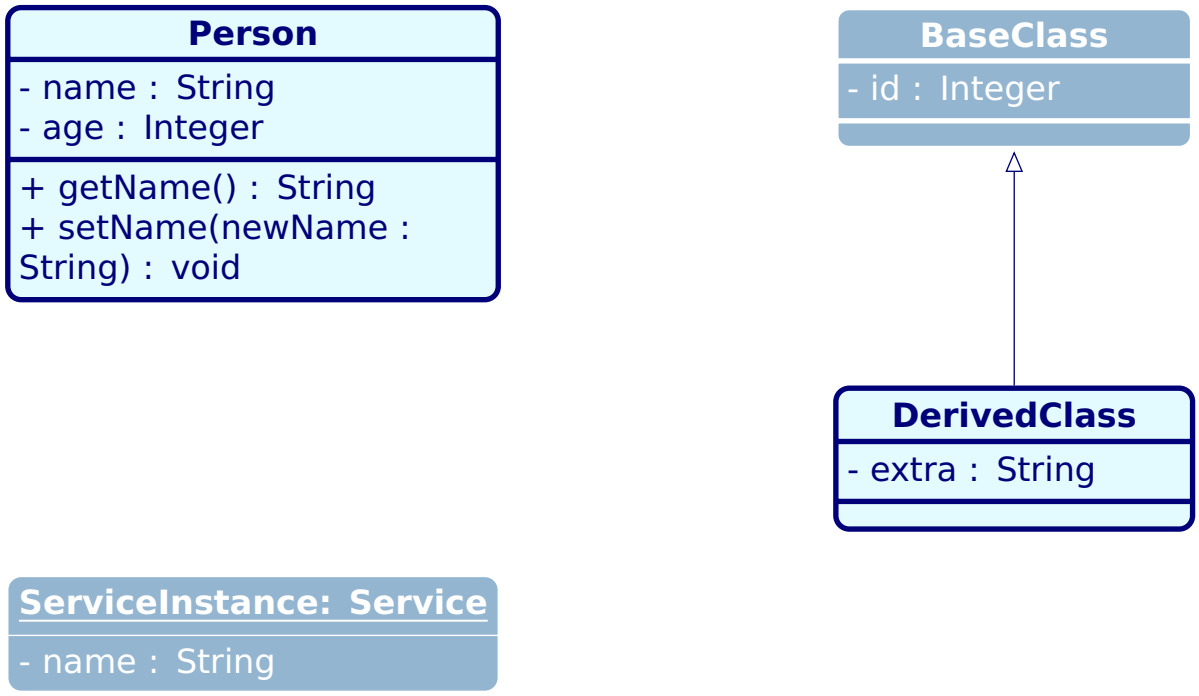
ON, ON.
Scalar SCALAR.flag: true.
Scalar SCALAR.text: Hello, world..
Missing key: [].

2.6 List clear and compact API

Length before clear: 3. Length after clear: 0. Values: x,y.
After `\pxSETlst`: z,a. After `\pxSET`: A. After `\pxPUT`: AB. With separator: A, B.

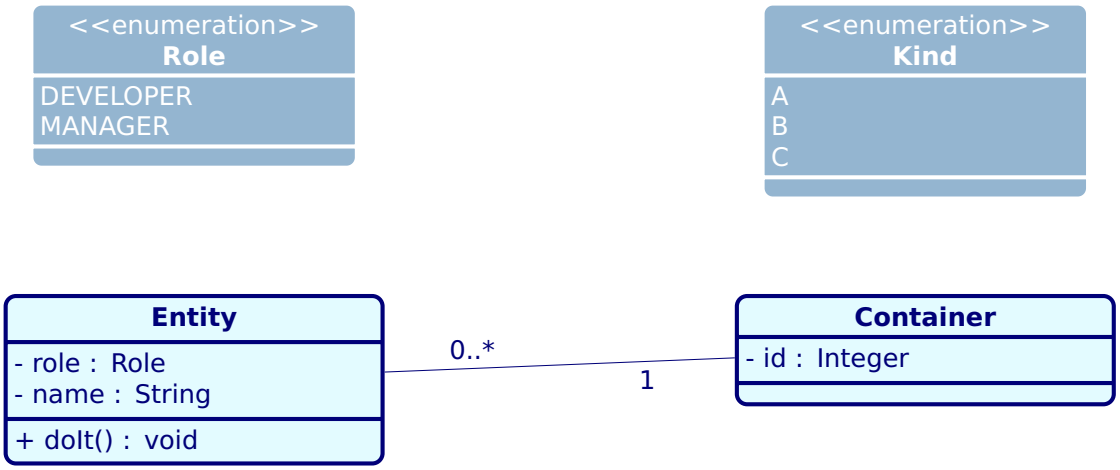
3 pxUML: class diagrams

3.1 Classes and diagram



Inspection: Person attributes: [- name : String,- age : Integer].
Service operations: [+ run() : void].
Parents of DerivedClass: [BaseClass].
Non-existent class: [].

3.2 Enums and associations



4 pxSRC: safe images

Figure 1: Single image caption

Figure 2: Caption (optional width; color uses default)

Figure 3: Pair caption

Figure 4: Step one

Figure 5: Step two

Figure 6: Step three

Figure 7: Step four

5 pxTAB: predefined tables

Name	Age	City
Alice	20	Barcelona
Bob	22	Madrid

Item	Quantity
Apples	3
Oranges	5
Bananas	2

Code	Description
PX1	First code
PX2	Second code

m11	m12	m13
m21	m22	m23

Language	Users
LaTeX	Many
P3CTeX	Felizment cada cop més

Table 1: Sample pxTAB float table

Module	Feature
pxSRC	Safe image inclusion
pxTAB	Predefined table styles

Table 1 shows a small pxTAB float that can be cross-referenced in the usual way.

Criterion	Weight	Notes
Design	40%	Structure and clarity
Implementation	40%	Correctness and style
Testing	20%	Coverage and edge cases

Student	PEC 1	PEC 2	Final
Alice	8.0	9.0	8.5
Bruno	7.5	8.0	7.8
Carla	9.0	9.5	9.3
Daniel	6.5	7.0	6.8

Component	Weight	Description
Design	30%	Justify architectural decisions and document the chosen patterns. Include a class diagram and a brief rationale for each design trade-off.
Implementation	40%	Deliver clean and well-structured source code with meaningful variable names. The build must succeed without errors or critical warnings.
Testing	20%	Provide unit tests covering the main public API and at least two edge cases per module.
Report	10%	Summarise the work in a concise document using the P3CTeX template. Include references and an appendix with test logs.

Module	Credits	Grade
Algebra	6	A
Calculus	6	B+

Component	Status
API Design	Complete
Implementation	In progress

Test	Result
Unit	Pass
Integration	Pass

Name	Course	Grade
Alice	CS101	A
Bob	CS102	B+

Columna 1	Columna 2	Columna 3
11	12	13
21	22	23

Columna 1	**Columna 2**	**Columna 3**
31	32	33

1	2	3	4	5	6	7	8
X	Y	Z					
X	Y	Z					
X	Y	Z					

1	2	3	4	5	6	7	8
P	A	T	A	T	A		
P	A	T	A	T	O	D	OS

6 pxCORE: setup and debug

Debug state: [DEBUG-OFF].

7 pxGDX: layout helpers

Example title text

8 pxLST: listats de codi professional

lst-java-ex

```
public class Fibonacci {
    public static int fib(int n) {
        if (n <= 1) return n;
        int a = 0, b = 1;
        for (int i = 2; i <= n; i++) {
            int tmp = a + b;
            a = b; b = tmp;
        }
        return b;
    }
}
```

codebox:[lst:alg-ex] Algorisme: factorial

```
Funcio factorial(n: Integer): Integer
  Si n ≤ 0 Llavors
    Retornar 1
  Sinó
    resultat ← 1
    Per i = 1 A n Fer
      resultat ← resultat × i
    Fi Per
    Retornar resultat
  Fi Si
Fi Funcio
```

codebox:[lst-alg-ex]

La llista 8 mostra l'algorisme iteratiu del factorial.

lst-yaml-ex

```yaml
project:
  name: P3CTeX
  version: "0.3"
  languages: [ca, en]
  debug: false
```

lst-bash-ex

```bash
#!/bin/bash
cd tex/
pdflatex -halt-on-error main.tex
echo "Build complet."
```